Research Report

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Introduction

Electric vehicle (EV) prices vary substantially across models, even among vehicles with similar technical specifications. Battery capacity is one of the most visible EV characteristics and is often associated with both production cost and consumer value. However, vehicle price is not determined by battery size alone. Manufacturers differ in brand positioning, and premium brands may charge systematically higher prices than non-premium brands even for vehicles with similar specifications.

This report studies EV models listed for sale in Germany and investigates whether battery capacity is associated with vehicle price, and whether that association differs between premium and non-premium manufacturers. The research questions are:

1. Among electric vehicle models listed for sale in Germany, is battery capacity positively associated with vehicle price?
2. Does the association between battery capacity and price differ between premium and non-premium manufacturers?

The data come from the *Electric Vehicle Specifications and Prices* dataset obtained from Kaggle and compiled by web scraping EV Database. EV Database aggregates manufacturer-reported technical specifications and listed prices for electric vehicle models listed for sale in Germany. The Kaggle dataset page indicates that the dataset entry was updated "2 years ago," which provides some background on when the posted dataset was last refreshed. However, that platform update note should not be interpreted as a precise statement of the model year coverage or the exact scrape date of the underlying EV listings. The dataset contains 360 observations, where each row represents a distinct EV model. This dataset should not be interpreted as a complete census of all electric vehicles that have ever existed or all EVs ever sold in Germany. Instead, it is best understood as a snapshot of the EV models included in EV Database when the source was scraped and the Kaggle dataset was assembled. The exact calendar year or scrape date is not stated in the raw CSV used here, so the report avoids treating the data as representing a single fully specified year. Since the data are observational rather than experimental, all conclusions in this report are associative rather than causal.

The response variable is `Price.DE.`, the listed vehicle price in euros. The main explanatory variable is `Battery`, measured in kilowatt-hours (kWh). A second key variable is manufacturer segment, coded as premium or non-premium. Following the proposal, Tesla, BMW, Audi, Mercedes-Benz, and Porsche are classified as premium brands based on their widely recognized positioning in the European automotive market. This grouping is intended to reflect broad brand-tier differences rather than a formal regulatory classification. Other quantitative variables that may be useful during model building include `Range`, `Efficiency`, `Top_speed`, `Acceleration..0.100.`, and `Fast_charge`.

This report uses multiple linear regression because the response is continuous and the main goal is inference about average relationships. In particular, including an interaction between battery capacity and market segment allows the battery-price relationship to differ between premium and non-premium manufacturers, which directly addresses the second research question. Based on the course material, variables named in the research question remain in the model throughout model selection even if a selection procedure would otherwise remove them.

Exploratory Analysis

Data Preparation

The raw dataset is stored locally in the project folder as `data/EV_cars.csv`. Using a relative file path makes the analysis reproducible for all group members and avoids machine-specific paths such as `~/Desktop/...`.

The raw file contains 360 observations and 9 recorded variables: `Battery`, `Car_name`, `Car_name_link`, `Efficiency`, `Fast_charge`, `Price.DE.`, `Range`, `Top_speed`, and `acceleration..0.100.`. Because the data were scraped from an online database, the observed models reflect what EV Database included during the underlying collection period, not necessarily every EV available in the German market before or after that time. This matters for interpretation: the analysis is best understood as describing relationships among the EV models represented in this snapshot, rather than all EVs from a specific stated year. The response variable `Price.DE.` has 51 missing values, and `Fast_charge` has 2 missing values. Since price is the response, rows with missing price cannot be used for the regression analysis and are removed. The two rows with missing `Fast_charge` are retained during initial cleaning and are only excluded in models that include `Fast_charge`.

To prepare the analysis dataset, a manufacturer variable was extracted from the vehicle name, the premium/non-premium grouping from the proposal was applied, the acceleration variable was renamed to a cleaner label, and only the variables relevant to the analysis were retained. Identifier variables such as `Car_name_link` were not used as predictors because they do not have an interpretable statistical relationship with price.

After removing rows with missing price, the main analysis dataset contains 309 EV models. A check of duplicate `Car_name` values shows that some names appear more than once. These rows were retained because they are not exact duplicates of the full record. In several cases, the repeated names correspond to different EV Database links and slightly different technical specifications, which suggests updated entries or distinct configurations rather than accidental duplication. For example, duplicated Mercedes entries often appear as an older version with missing price and a newer version with a reported price, while the duplicated Mini Cooper SE entries differ substantially in battery size and range and therefore represent genuinely different configurations.

For the main regression, one practical choice is to fit models using complete cases on the variables included in each model. If `Fast_charge` remains in a candidate model, the effective sample size becomes 307 rather than 309 because of the two missing charging values. This should be noted when comparing models. The 51 models with missing prices do not appear dramatically different from the retained models in terms of battery capacity, range, efficiency, top speed, acceleration, or fast-charge capability, which reduces concern that the missing prices are concentrated only among very low-end or very high-end vehicles. However, because the missingness mechanism is not known, some selection bias remains possible.

Exploratory Summaries

Exploratory analysis serves two purposes in this study. First, it helps describe the data to readers who are not familiar with the EV market. Second, it helps identify features of the data that affect the regression analysis, such as skewness, segment differences, and multicollinearity among technical variables.

Start with summary statistics for the response and candidate explanatory variables.

Table 1: Summary statistics for key quantitative variables

N	Mean price	Median price	SD price	Mean battery	Median battery	SD battery	Mean range	Mean efficiency
309	67264.28	56942	34508.25	71.08	70.5	20.65	369.13	195.41

Among the 309 EV models used in the main analysis, the mean listed price is about 67,264 EUR and the median price is 56,942 EUR, indicating some right-skew from a smaller number of high-priced vehicles.

Battery capacity has a mean of 71.1 kWh and a median of 70.5 kWh, so the battery distribution is more balanced than the price distribution. When the sample is split by manufacturer segment, premium vehicles are clearly larger and more expensive on average: premium models have a mean price of about 94,359 EUR and a mean battery size of 84.0 kWh, compared with 55,418 EUR and 65.4 kWh for non-premium models.

The summary statistics already suggest two important features that matter for modeling: price is more variable and somewhat right-skewed, while premium vehicles tend to have both larger batteries and higher prices. This motivates focusing on the joint relationship between price, battery capacity, and market segment.

The most important exploratory plot is the scatterplot of price against battery capacity, with separate fitted lines for premium and non-premium manufacturers. This directly addresses both research questions.

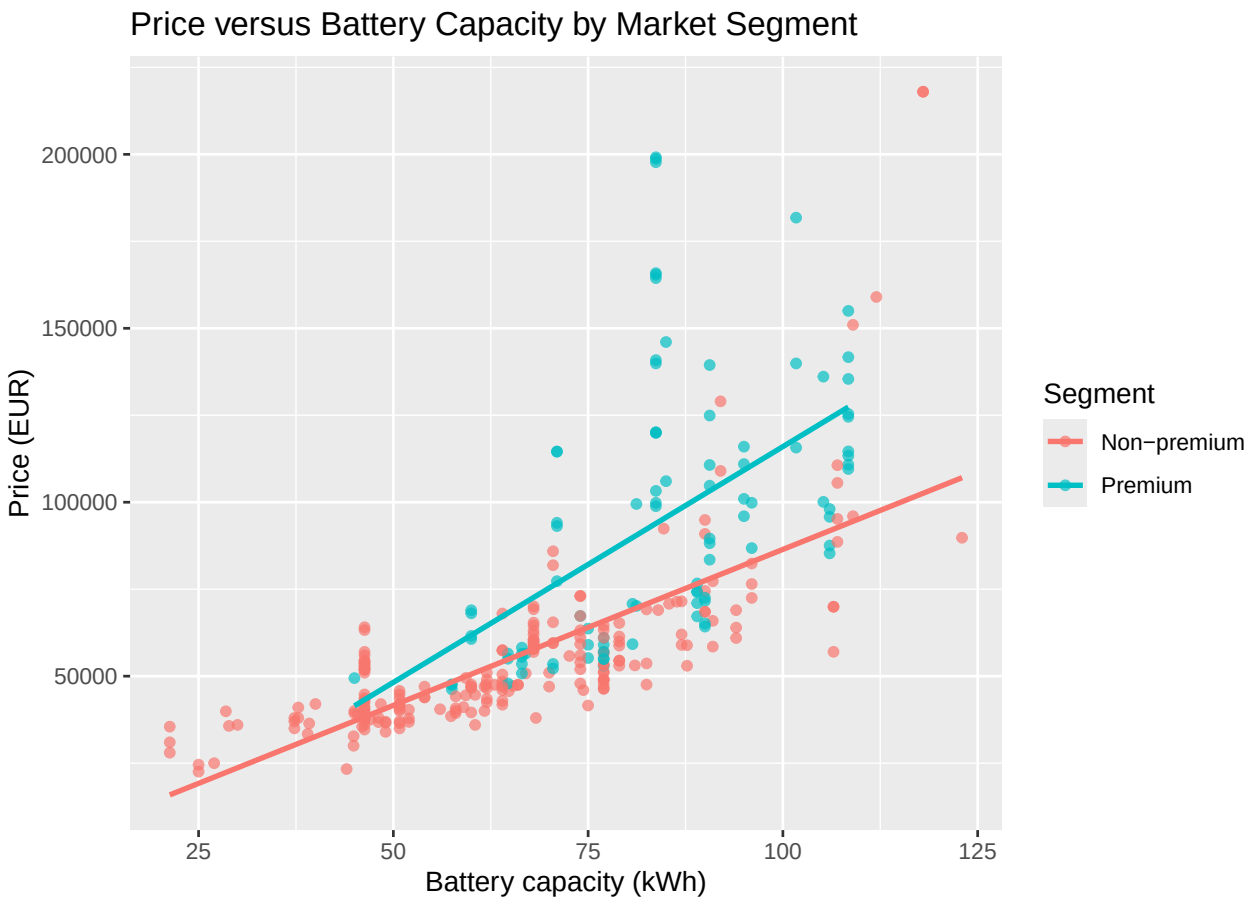

Figure 1: Price versus battery capacity by market segment

The scatterplot indicates a clear positive association between battery capacity and price overall. It also shows strong separation between the two market segments. Premium vehicles tend to appear both higher and farther to the right on the plot, meaning that they generally have larger batteries and higher prices. Simple fitted lines within each segment are positive, with an estimated unadjusted increase of about 897 EUR per kWh for non-premium vehicles and about 1,355 EUR per kWh for premium vehicles. This visual pattern suggests that the battery-price relationship may indeed differ by manufacturer segment.

Because several quantitative vehicle characteristics may be related to one another, it is also useful to examine pairwise relationships and correlations. This can help identify potential multicollinearity before model fitting.

Table 2: Correlation matrix for quantitative variables

	Price.DE.	Battery	Range	Efficiency	Top_speed	Acceleration_0_100	Fast_charge
Price.DE.	1.00	0.70	0.59	0.17	0.76	-0.53	0.62
Battery	0.70	1.00	0.88	0.16	0.70	-0.61	0.59
Range	0.59	0.88	1.00	-0.31	0.76	-0.75	0.72
Efficiency	0.17	0.16	-0.31	1.00	-0.21	0.42	-0.29
Top_speed	0.76	0.70	0.76	-0.21	1.00	-0.84	0.79
Acceleration_0_100	-0.53	-0.61	-0.75	0.42	-0.84	1.00	-0.68
Fast_charge	0.62	0.59	0.72	-0.29	0.79	-0.68	1.00

The correlation matrix also highlights why model selection is needed. Price is strongly positively correlated with `Battery` (0.70) and `Top_speed` (0.76), while `Battery` and `Range` are themselves highly correlated (0.88). `Top_speed` and `Acceleration_0_100` are also strongly associated in magnitude, with a correlation of -0.84 because faster vehicles have lower acceleration times. These patterns suggest substantial overlap among the technical variables, so a simpler regression model is preferable for interpretation.

Statistical Analysis

Modeling Strategy

The response is continuous, so multiple linear regression is appropriate. The variables appearing explicitly in the research question, `Battery`, `market_segment`, and their interaction, remain in all candidate models. Additional quantitative variables are treated as optional controls and may be removed during model selection if they do not meaningfully improve the model.

Following the course guidance for inference-focused projects, the main selection approach used here is backward elimination based on partial F-tests for optional variables only, while keeping the research-question terms fixed in the model. A test-based method was preferred because the goal of the study is inference rather than prediction. This gives a simpler model without dropping the variables needed to answer the main questions of interest.

Initial Model

The coefficient of `Battery` estimates the battery-price association for the baseline group, which is non-premium manufacturers. The coefficient of `market_segmentPremium` estimates the difference in intercept between premium and non-premium manufacturers. The interaction term `Battery:market_segmentPremium` estimates how much the battery-price slope differs for premium brands relative to non-premium brands.

The initial model includes the research-question terms and the additional quantitative variables considered plausible controls: `Range`, `Efficiency`, `Top_speed`, `Acceleration_0_100`, and `Fast_charge`. This serves as the starting point for the test-based model selection procedure.

Model Selection

Backward elimination was implemented using partial F-tests through `drop1(..., test = "F")`. Only optional variables were considered for removal, while the three terms tied directly to the research questions were retained throughout the selection process.

Backward selection removed `Range` and `Acceleration_0_100`, while retaining `Efficiency`, `Top_speed`, and `Fast_charge` in addition to the required research-question terms. This matches the exploratory evidence of multicollinearity: once `Battery`, `Top_speed`, and the other retained controls are in the model, `Range` no longer provides much additional information. By contrast, `Top_speed` was retained because the selection results suggest that it provides additional explanatory power for price beyond what is already captured by

battery capacity and the other predictors. The selected model keeps the key interaction term and remains highly explanatory, with an adjusted R^2 of about 0.729.

Main Model Results

The selected model removes `Range` and `Acceleration_0_100` while retaining `Efficiency`, `Top_speed`, and `Fast_charge` along with the required interaction structure. This is the main model used for interpretation because it is simpler than the full model and fits nearly as well.

Table 3: Coefficient estimates for the final regression model

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	-119258.2125	10496.8459	-11.3613	0.0000
Battery	63.4378	86.8515	0.7304	0.4657
market_segmentPremium	-41215.8559	11502.2868	-3.5833	0.0004
Efficiency	335.9446	37.8674	8.8716	0.0000
Top_speed	565.3966	56.0359	10.0899	0.0000
Fast_charge	20.2689	7.3438	2.7600	0.0061
Battery:market_segmentPremium	597.9874	138.2935	4.3240	0.0000

In the selected model, the baseline group is non-premium vehicles. The estimated coefficient for `Battery` in that group is about 63 EUR per kWh and is not statistically significant on its own ($p = 0.466$). The interaction term represents the difference in battery-price slope between premium and non-premium vehicles, and its estimate is about 598 EUR per kWh with $p < 0.001$. The estimated battery-price slope for premium vehicles is therefore $63 + 598 \approx 661$ EUR per kWh. These results provide strong evidence that the battery-price relationship differs between premium and non-premium manufacturers after controlling for the retained technical variables. The intercept is negative, but it has no practical interpretation because no vehicle in the dataset has zero battery capacity; it is a mathematical artifact of the linear model. The coefficient on `market_segmentPremium` by itself is also defined at a battery size of zero and is therefore not substantively meaningful on its own. The more relevant interpretation comes from the interaction and the fitted relationship: premium vehicles are priced differently and their prices rise more steeply with battery capacity. Although the non-premium battery coefficient is small relative to the premium slope and is not statistically significant, its practical meaning is still worth considering: for example, a 40 kWh increase would correspond to about 2,520 EUR under this estimate. One plausible explanation for the weak statistical evidence is that prices for non-premium vehicles may be driven more by standardized model tiers and bundled performance features than by battery capacity alone, or that the within-segment relationship is estimated with greater variability once the other technical controls are included. The lack of significance of `Battery` in the non-premium group may also suggest that battery capacity is partly reflected through other retained performance-related variables, especially `Top_speed`.

Diagnostics and Sensitivity Check

After selecting the final model, standard diagnostic checks should be performed. These checks help assess linearity, constant variance, approximate normality of residuals, and influential observations.

If the residual plots show strong heteroscedasticity or right-skewed residuals, a log transformation of price may be worth considering. The transformed model should only be adopted if diagnostics clearly improve and the interpretation remains appropriate for the study objective.

Diagnostic plots were examined for nonlinearity, non-constant variance, and influential observations. The residual-versus-fitted plot did not show strong curvature, and the normal Q-Q plot did not indicate severe departures from normality. However, the spread of residuals appears to increase somewhat for larger fitted values, which suggests mild heteroscedasticity. Supporting diagnostic plots are provided in the appendix. For this reason, a log-price model was considered as a secondary check. The log-price model fits somewhat

better, with an adjusted R^2 of about 0.841, and the battery-by-segment interaction remains positive and highly significant. More importantly, the substantive conclusion is the same in both models: the battery-price relationship is much stronger for premium vehicles than for non-premium vehicles. Although the log-price model improves fit and likely reduces heteroscedasticity, the untransformed model is retained as the primary model because its coefficients can be interpreted directly in EUR, which is more meaningful for addressing the research questions.

Discussion and Conclusion

The exploratory plots and the regression analysis both point to the same broad conclusion: the relationship between EV price and battery capacity depends on manufacturer segment. In the raw scatterplot, both segments show upward trends, and premium vehicles tend to occupy the higher-price, larger-battery portion of the figure. After controlling for efficiency, top speed, and fast-charging capability, the selected regression model shows that the battery-price slope is much steeper for premium brands than for non-premium brands.

For the first research question, the evidence suggests that the positive battery-price association is driven primarily by premium vehicles. After adjustment for the retained controls, each additional kWh of battery is associated with only about 63 EUR more in price for a non-premium vehicle, and this effect is not statistically significant. For a premium vehicle, the corresponding estimated increase is about 661 EUR per kWh. For the second research question, the interaction term provides strong evidence that the battery-price relationship differs by segment. This suggests that battery capacity is valued differently across market segments, consistent with premium brands using larger batteries as part of a broader high-end product strategy.

The retained control variables also help clarify the pricing structure. `Top_speed`, `Efficiency`, and `Fast_charge` remain statistically important in the selected model, indicating that price reflects a bundle of performance and charging characteristics rather than battery size alone. By contrast, `Range` and `Acceleration_0_100` do not provide enough additional information once the retained variables are included, which is consistent with the strong correlations observed during the exploratory analysis. More broadly, the lack of a strong battery effect in the non-premium baseline group may indicate that battery capacity is partly associated with price through other performance-related characteristics captured by the retained predictors.

Several limitations remain important. The data are observational and based on public listings, so the results should not be interpreted causally. The dataset is also a snapshot of EV models included in EV Database at the time of collection rather than a complete record of all EVs across all time periods, and the exact calendar year represented by the snapshot is not explicitly stated in the raw file. As a result, the findings apply most directly to the models represented in this dataset rather than to all EVs in a precisely defined year. The premium/non-premium classification is a simplified representation of brand positioning, and some important determinants of price, such as trim level, software features, interior quality, and brand-specific technology, are not directly measured. In addition, the analysis is specific to vehicles listed in Germany and may not generalize to other markets.

Overall, the analysis supports the conclusion that battery capacity is related to EV price, but not in the same way for all manufacturers. Premium brands appear to attach a stronger price premium to additional battery capacity, and this difference remains visible even after accounting for other technical characteristics.

Statement of AI Use

Generative AI was used to help organize the structure and wording of this report and to assist in preparing a separate reproducible analysis script. The statistical conclusions, numerical results, and model interpretations are based on outputs generated from the dataset itself.

Appendix

Notes on Course Alignment

The methods used in the main report are limited to those covered in the course and are chosen because they directly address the data and research questions:

- exploratory summaries and a scatterplot with fitted lines
- multiple linear regression with an interaction between a quantitative and a categorical variable
- backward model selection based on partial F-tests for optional variables
- standard residual diagnostics for the chosen model
- an optional log transformation of the response as a secondary diagnostic check

Additional Supporting Details

The main body reports only the outputs most relevant to the research questions in order to stay concise. Additional steps used in the analysis include:

- checking missingness and duplicate model names during data preparation
- fitting the initial full model before model selection
- examining residual and influence diagnostics
- fitting a log-price sensitivity model to assess whether transformation changes the main conclusions

These supporting analyses inform the final model choice, but the key findings and interpretations are presented in the main body of the report.

Diagnostic Plots

The following plots provide supporting evidence for the model-diagnostic statements made in the statistical analysis section.

Pairwise Relationships Among Quantitative Variables

As an additional exploratory check, the following scatterplot matrix summarizes pairwise relationships among the quantitative variables used in the analysis. This is included in the appendix rather than the main body because it is useful for supporting model-building decisions, but it is not the most direct figure for answering the research questions.

This plot is mainly helpful for visually checking which explanatory variables appear strongly associated with price and which predictors appear highly related to one another. In this dataset, it supports the conclusion that variables such as `Battery`, `Range`, `Top_speed`, and `Fast_charge` share substantial overlap, which helps explain why a simpler final model is preferred.

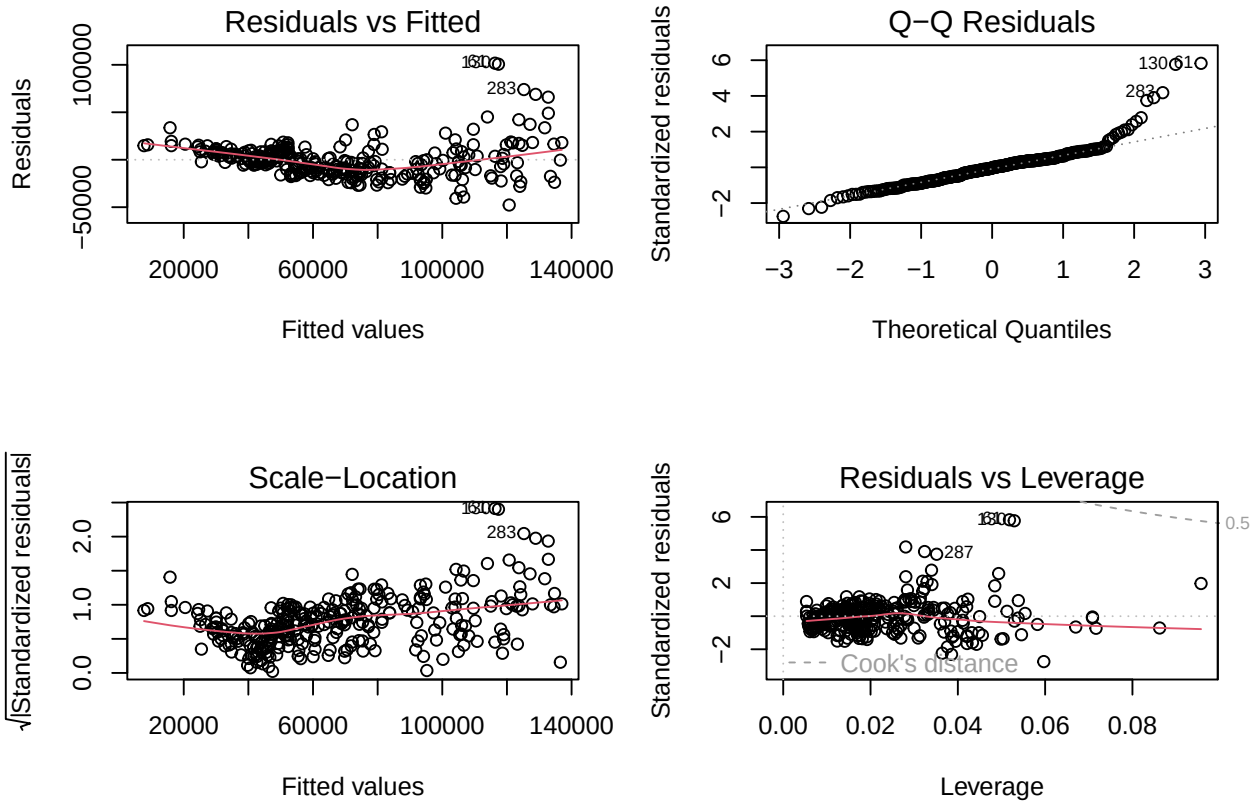

Figure 2: Diagnostic plots for the selected regression model

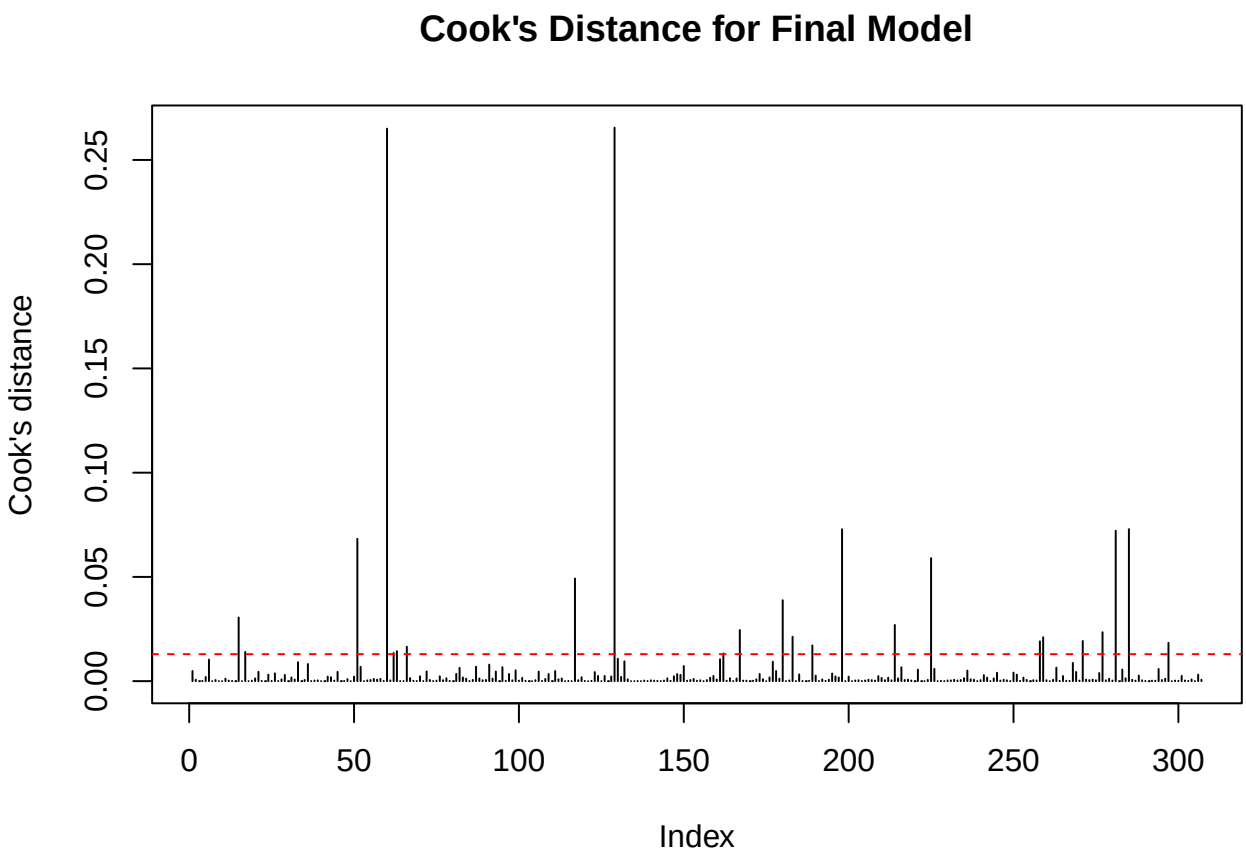

Figure 3: Cook's distance for the selected regression model

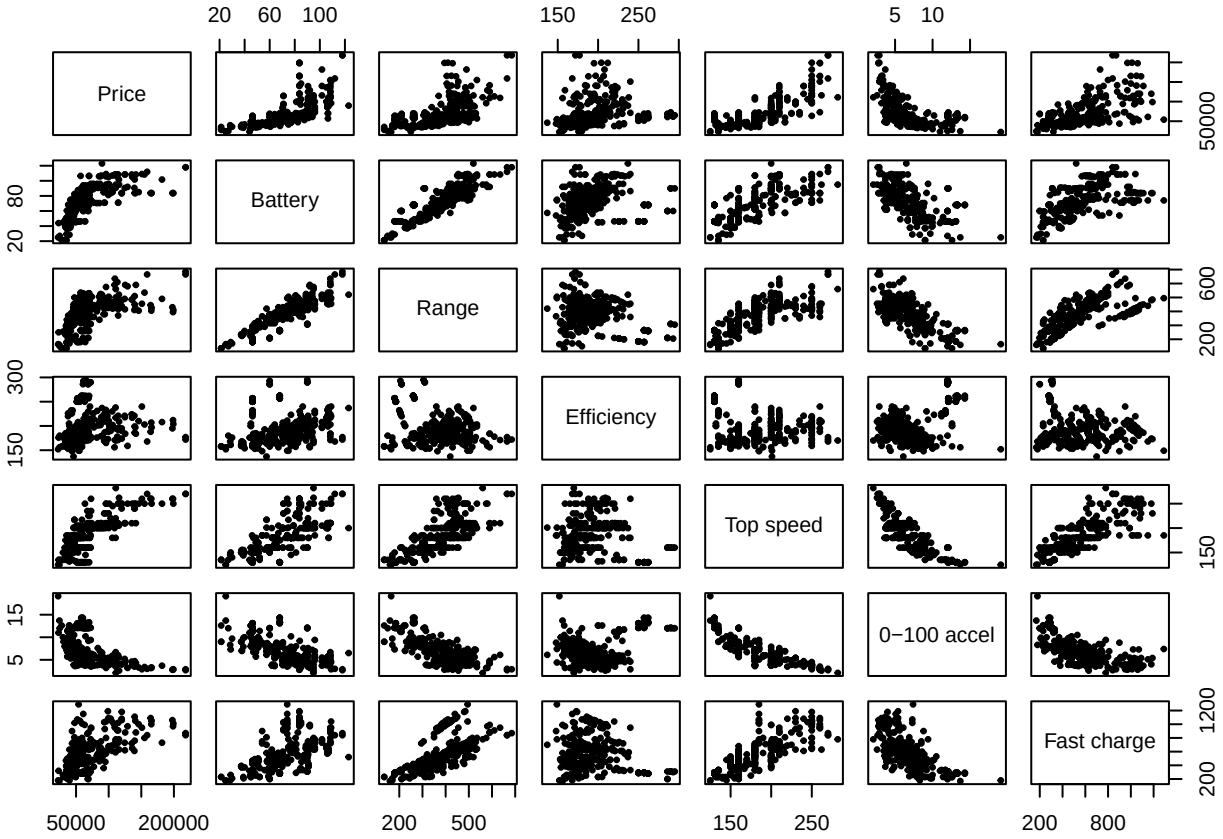

Figure 4: Scatterplot matrix for quantitative variables used in the analysis